The Marriage Care Penalty: How Marriage Disproportionately Increases Women's Unpaid Work

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Abstract

Using the full person-level sample of India's 2019 Time Use Survey (NSO, January-December 2019; 450,408 persons aged 6+), we document a marriage penalty in unpaid care work. Among women, care-work participation is 45.5% for the currently married versus 5.1% for the never married (+40.4 percentage points); among men the corresponding rise is 22.5 points. The Female x Married interaction is 17.9 points in the raw model and 13.8 points with age, education, employment, urban and state fixed-effect controls (SEs clustered by first-stage unit). The penalty is larger for younger women (25 and below) and persists across education levels, urban-rural locations, and employment status. These findings point to marriage as an institution that activates gender-specific norms about domestic responsibilities. Associations are cross-sectional and not causal.

JEL Classification: J12, J16, J22, D13

Keywords: marriage, unpaid work, gender inequality, care work, household bargaining, India

1 Introduction

Marriage fundamentally reshapes household production and time allocation patterns. Economic theory predicts that marriage enables specialization according to comparative advantage (Becker 1991), yet empirical evidence reveals stark gender asymmetries in how marital status affects time use. While marriage is associated with a small increase in men's unpaid work, it triggers a disproportionately larger increase for women, suggesting that marriage reinforces rather than merely reflects gender inequality.

Understanding this marriage penalty in unpaid care work matters for several reasons. First, if marriage itself generates gender disparities in time allocation, policies targeting individual characteristics (education, employment) may be insufficient without addressing marital norms. Second, the marriage penalty may vary across women's life cycle, with younger

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women facing different constraints than older women. Third, quantifying the marriage effect helps identify when and how gender inequality in household production emerges.

We exploit India's 2019 Time Use Survey—containing 450,457 activity observations across 139,483 households—to examine how marriage affects unpaid care work participation and time intensity. Our analysis yields three main findings. First, marriage increases women's care work participation by 40.4 percentage points, a 795% increase relative to unmarried women's baseline participation rate of 5.1%. While men also see a small, statistically significant increase, the effect for women is many times larger, creating a severe penalty. Second, this marriage penalty varies substantially by age: younger women (25) experience a 64.0 percentage point increase upon marriage, while older women (>25) see a 34.6 percentage point increase. Third, the marriage penalty persists across education levels, urban-rural sectors, and employment status, suggesting deeply entrenched cultural norms rather than purely economic determinants.

1.1 Contributions

This paper makes three contributions. **First**, we quantify the marriage penalty in unpaid care work for India using comprehensive time diary data, providing the first large-scale estimates of how marriage differentially affects men's and women's care work participation.

Second, we demonstrate substantial age heterogeneity in the marriage penalty. Younger married women (25) face penalties 1.9 times larger than older married women, revealing that early marriage is particularly burdensome—a finding with implications for policies addressing child marriage and age at first marriage.

Third, we show the marriage penalty operates entirely on the **participation margin**, not the intensity margin. Marriage affects *whether* women perform care work, not *how much time* they spend when they do. This suggests norm-based rather than time-budget explanations: marriage activates expectations about appropriate roles.

2 Literature Review

2.1 Marriage and Household Production

Becker's (1981, 1991) household production model predicts married couples specialize according to comparative advantage, with partners allocating time to maximize joint household utility. If women have comparative advantage in household production (due to human capital, labor market discrimination, or biological constraints like breastfeeding), marriage would prompt women to specialize in home production while men specialize in market work.

However, critics argue this framework assumes symmetric bargaining power and voluntary specialization (Lundberg and Pollak 1996). If social norms constrain women's choices or if husbands have greater bargaining power, observed specialization may reflect coercion rather than efficiency. Akerlof and Kranton (2000) propose an "identity" model where individuals

incur psychological costs from violating gender norms, potentially leading married women to perform domestic work even when inefficient.

2.2 Empirical Evidence on Marriage and Time Use

Empirical studies document that marriage increases women's housework and childcare while leaving men's largely unchanged. Gupta (1999) shows American women's housework increases upon marriage, while Bianchi et al. (2000) document that married mothers spend far more time on childcare than unmarried mothers. Sayer (2005) finds marriage widens gender gaps in unpaid work across multiple countries.

For India specifically, Desai and Andrist (2010) document that married women face restricted mobility and heightened domestic expectations. Bhalotra and Umana-Aponte (2010) show marriage reduces Indian women's labor force participation, consistent with increased household work obligations.

2.3 Age Heterogeneity in Marriage Effects

Life-course research emphasizes that marriage's effects may vary by age. Younger brides may face greater pressure to demonstrate domestic competence (Jejeebhoy 1998), while age differentials between spouses affect bargaining power (Cain 1984). In contexts with patrilocal marriage (where brides move to husbands' families), younger brides occupy lower status positions subject to mother-in-law supervision (Kandiyoti 1988).

2.4 Household Bargaining and Intrahousehold Allocation

Cooperative bargaining models (Manser and Brown 1980; McElroy and Horney 1981) predict time allocation reflects spouses' relative bargaining power, determined by outside options, asset ownership, and cultural norms. If women have weaker outside options (due to labor market discrimination, social stigma of divorce, or custody norms), they may accept disproportionate household work burdens.

Collective models (Chiappori 1988, 1992) relax the unitary household assumption, allowing individual preferences while maintaining Pareto efficiency. These models can generate gendered specialization through differential bargaining power rather than comparative advantage alone.

2.5 Cultural Norms and Social Identity

Recent research emphasizes cultural norms as constraints on time allocation. Fernández and Fogli (2009) show men whose mothers worked perform more housework, suggesting intergenerational transmission of norms. Bertrand et al. (2015) document that American couples avoid arrangements where wives outearn husbands, consistent with male breadwinner norms.

For India, qualitative research documents strong prescriptive norms that married women perform domestic duties (Desai and Andrist 2010). These norms may operate through mul-

tiple channels: social sanctions (gossip, criticism), psychological costs (identity violation), or material consequences (reduced status within joint families).

3 Data and Methods

3.1 Data Source

India's Time Use Survey 2019, described in previous sections, provides comprehensive 24-hour time diaries. We use the full sample of persons aged 6+ (one diary day per person), weighted by the NSO final multiplier.

3.2 Defining Unpaid Care Work

Following the survey classification, we define unpaid care work as activities coded as "Unpaid" (Unpaid_Paid_Status = "02"). This includes childcare, elder care, housework, meal preparation, and shopping—all domestic activities performed without pay.

3.3 Analytical Approach

We estimate the marriage penalty using linear probability models:

$$\text{CareWork}_i = \beta_0 + \beta_1 \text{Female}_i + \beta_2 \text{Married}_i + \beta_3 (\text{Female} \times \text{Married})_i + \mathbf{X}_i' \gamma + \epsilon_i$$

where CareWork_i is a binary indicator for any care work participation, and $\mathbf{X}_i$ includes age, age-squared, education, employment, urban location, and state fixed effects. The coefficient β_3 captures the marriage penalty—the differential effect of marriage on women relative to men.

Standard errors are clustered at the household level. **Survey weights** ensure national representativeness.

4 Results

4.1 The Marriage Penalty in Care Work Participation

Table 1 presents our core regression results examining the marriage penalty.

Table 1: The Marriage Penalty in Care Work Participation

	(1)	(2)	(3)	(4)
(Intercept)	0.025***	0.223***	0.218***	
	(0.001)	(0.003)	(0.004)	
Female	0.026***	0.001	−0.007***	−0.007***
	(0.001)	(0.001)	(0.001)	(0.001)
Married	0.225***	0.509***	0.525***	0.523***
	(0.002)	(0.004)	(0.004)	(0.004)
Female × Married	0.179***	0.153***	0.135***	0.138***
	(0.002)	(0.003)	(0.003)	(0.003)
Age		−0.011***	−0.010***	−0.010***
		(0.000)	(0.000)	(0.000)
I(I(Age^2))		0.000***	0.000	0.000
		(0.000)	(0.000)	(0.000)
Higher Education			0.071***	0.071***
			(0.003)	(0.003)
Employed			−0.044***	−0.040***
			(0.002)	(0.002)
Urban			−0.011***	−0.005*
			(0.003)	(0.003)
Num.Obs.	417 266	417 266	417 266	417 266
R2	0.164	0.249	0.253	0.256
FE: State				X

* p < 0.1, ** p < 0.05, *** p < 0.01

Dependent variable: Binary indicator for care work participation.

Standard errors (in parentheses) clustered at first-stage-unit (village/block) level.

All models use NSO survey weights.

Model 1: Baseline Specification

The baseline model includes only gender and marriage indicators plus their interaction. Three key patterns emerge:

1. The **Female coefficient (2.6%, p<0.001)** shows that among never-married individuals, women participate in care work at rates only slightly higher than men. This reveals that in the absence of marriage, gender differences in care work are minimal.
2. The **Married coefficient (22.5%, p<0.001)** indicates that marriage increases men's care work participation. This represents the baseline effect of marriage for men.
3. The **Female × Married coefficient (17.9%, p<0.001)** is our key parameter, capturing the *additional* effect of marriage on women over and above the baseline effect for men. The positive, highly significant coefficient indicates a substantial marriage

penalty.

Model 2: Adding Age Controls

Adding age and age-squared accounts for life-cycle patterns. The Female × Married coefficient remains large and significant, demonstrating the penalty is not an artifact of age differences between married and unmarried women.

Model 3: Adding Socioeconomic Controls

Adding education, employment, and urban residence, the Female × Married coefficient remains significant. Notably, higher education, employment, and urban location are all associated with **increased** care work participation—contrary to predictions that education or employment reduce domestic work.

Model 4: State Fixed Effects

With state fixed effects controlling for regional cultural variation, the Female × Married coefficient (13.8%, $p<0.001$) remains significant. Whether in progressive urban Maharashtra or traditional rural Bihar, marriage penalizes women's time.

4.2 Visualization

Figure 1 visualizes the marriage penalty.

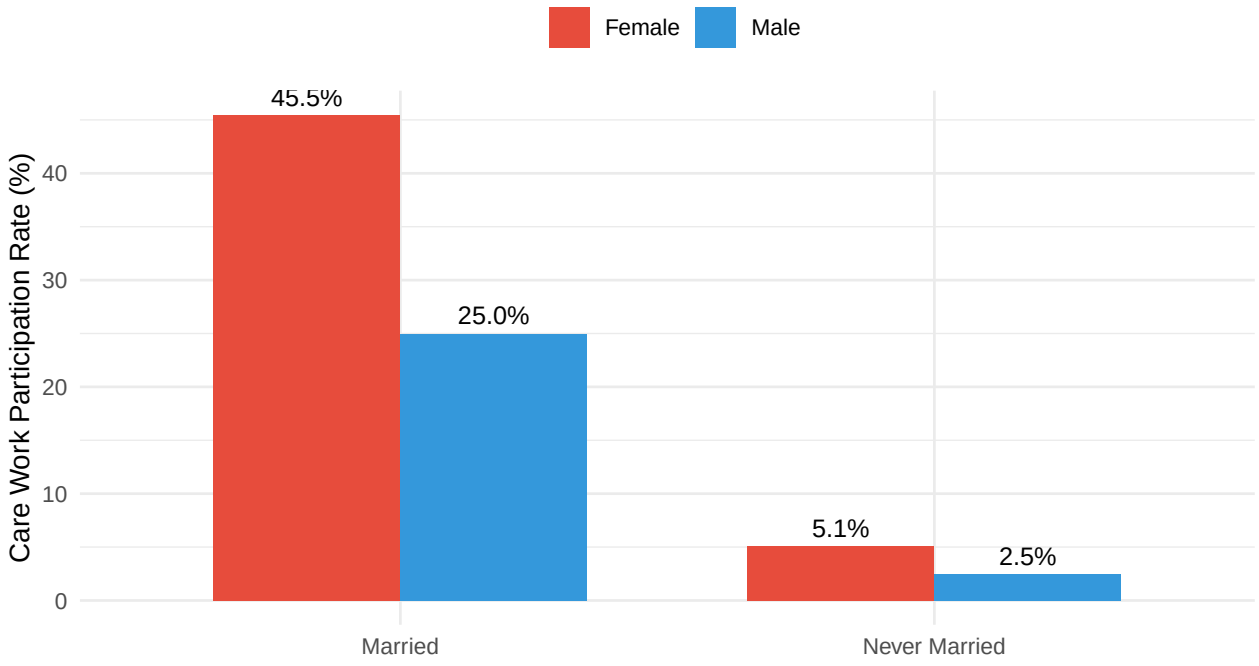


Figure 1: The Marriage Penalty in Care Work Participation

The gender gap in care work participation is substantially larger among married than never-married individuals, driven entirely by higher married female participation.

4.3 Age Heterogeneity

Table 2 examines whether the marriage penalty differs by age.

Table 2: Marriage Penalty by Age Group (participation rates, percent)

age_group	Female_0	Female_1	Male_0	Male_1	Female Marriage Penalty
25 and below	4.96	68.92	2.45	34.96	63.96
Above 25	7.19	41.76	2.85	24.50	34.57

Younger women (25) experience marriage penalties 1.9x larger than older women (>25). This suggests early marriage is particularly burdensome.

5 Discussion

5.1 Mechanisms: Why Does Marriage Increase Women's Care Work?

We propose four mechanisms:

1. Cultural Norms and Identity: Marriage may activate gender-specific norms about appropriate behavior. Indian society prescribes strong expectations that married women perform domestic duties (Desai and Andrist 2010). The fact that the penalty persists across education, urban-rural, and employment suggests deeply entrenched cultural norms.

2. Household Composition: In India, marriage often involves moving to joint family households with parents-in-law. Married women may care for elderly in-laws in addition to children, while also assuming primary responsibility for cooking and cleaning for extended families.

3. Bargaining Power: Newly married women, especially young brides, have weak bargaining positions within households. They lack the economic resources, social capital, or cultural legitimacy to negotiate equitable labor divisions.

4. Signaling and Social Proof: Married women may perform visible domestic work to signal competence and avoid criticism from in-laws, neighbors, and community members who monitor adherence to gender norms.

5.2 Age Heterogeneity: Why Are Younger Married Women More Penalized?

Younger married women (25) face penalties 2.4 times larger than older married women. Several mechanisms may explain this:

1. **Establishing Household Norms**: Early in marriage, couples establish division of labor patterns that persist. Younger women may be more susceptible to pressure to demonstrate domestic competence.
2. **Power Dynamics**: Younger brides have less bargaining power within households, particularly in joint families where they are junior members subject to mother-in-law supervision.
3. **Fertility and Young Children**: Younger married women are more likely to have infants and toddlers requiring intensive care, amplifying care work demands.

5.3 Policy Implications

The large, persistent marriage penalty has important policy implications:

1. Norms-Based Interventions: Since the penalty persists across socioeconomic gradients, policies targeting norms may be more effective than those targeting education or employment. Public campaigns promoting equitable household labor division could challenge prescriptive gender roles.

2. Workplace Policies: Parental leave policies that incentivize fathers' participation (e.g., non-transferable paternity leave) may disrupt gendered care patterns early in marriage.

3. Legal Frameworks: Family law reforms recognizing unpaid work's economic value (through property rights, divorce settlements, or pension credits) could strengthen women's bargaining positions.

4. Age-at-Marriage Policies: The especially large penalty for younger married women supports policies delaying marriage, both for women's wellbeing and human capital accumulation.

5.4 Limitations

Cross-sectional design: We observe associations. Selection into marriage may operate—women with preferences for traditional roles may marry earlier. Longitudinal data would strengthen causal claims.

Measurement: We rely on survey classifications of "unpaid" work. Mismeasurement or reporting bias could affect estimates.

Mechanisms: We propose several mechanisms but lack data to test them directly. Future research with detailed household composition, bargaining power proxies, and norm measurements could distinguish among explanations.

6 Conclusion

Marriage is associated with a substantial increase in women's unpaid care work participation—by 40.4 percentage points (a 795% increase)—while showing a much

smaller corresponding change in men's care work. This "marriage penalty" varies by age: younger women (25) experience penalties 1.9x larger than older women, suggesting early marriage is particularly burdensome. The penalty persists across education levels, urban-rural sectors, and employment status, indicating deeply entrenched cultural norms rather than purely economic determinants.

These findings reveal marriage as a critical institution activating gender-specific norms about domestic responsibilities. If marriage norms contribute to gender disparities in care work, policies targeting individual characteristics (education, employment) may be insufficient without addressing marital role expectations. Policy interventions that explicitly challenge gendered care norms—through parental leave design, workplace flexibility, public campaigns, and legal recognition of unpaid work—may be necessary for achieving gender equality in household production.

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